Planning of Rooftop Rainwater Harvesting Structure Based on GIS Approach

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ABSTRACT

The study was undertaken to estimate the roof water harvesting potential of Junagadh Agricultural University (Gujarat) campus using Google Earth and Quantum GIS and to evaluate the demand supply analysis. The total rooftop area of 81 buildings was found as 1,04,012 m^2. Total roof water harvesting potential of campus was estimated at 44426 kiloliter per year at design rainfall of 66% probability of exceedance and 77,889 kiloliter per year at design rainfall based on monthly average value. Sample calculation of demand supply analysis revealed that roof water harvesting potential at 66% probability of exceedance design rainfall can fulfill 78, 88, 59 and 100% yearly demand of college, hostel, multi storey office and residence *i.e.* up to 281st day, 292nd day, 212th day and 365th day of water year from 14844 m^2, 1110 m^2, 1036 m^2 and 294 m^2 rooftop area, respectively. For residence, 26.58 kiloliter additional water will be available for recharge. For monthly average design rainfall, 3016 kiloliter, 293 kiloliter, 27 kiloliter and 121.17 kiloliter water per year will be available for recharge for college, hostel, multi storey office and residence, respectively in addition to satisfying the yearly water demand. Maximum storage of water required at 66% probability of exceedance design rainfall for college, hostel, multi storey office and residence was found to be 3640.24, 297.7, 192.91 and 92.58 kiloliter, respectively while in case of monthly average design rainfall, it was 8415.93, 636.84, 526.21 and 187.17 kiloliter, respectively. The information on maximum storage required can be used for planning and cost estimation.

Keywords: *Water harvesting, Roof water harvesting, Design rainfall, Quantum GIS, Demand supply analysis*

INTRODUCTION

The climate change has caused an uneven spatial and temporal distribution of rainfall. In developing countries, rapid urbanization against limited resources has resulted in reduction of quality and quantity of fresh water availability due to over exploitation of ground water. The regions where rainfall is the only source of water for domestic, industrial and agricultural use, the situation is more alarming. Rooftop rainwater harvesting is a way to access least polluted source of water *i.e.* rainwater. Utilization of roof water can help to reduce the burden on other sources of water either by storage or ground water recharge (Paul *et al.*, 2016). Agarwal *et al.* (2010) stated that there is a need to adopt the roof water harvesting technique at mass level to get significant results to improve the quality and availability of water. The studies conducted by Adriano *et al.* (2011) in Philippines, Alam *et al.* (2012) in the rural areas of Bangladesh, Jankar and Bhanuse (2013) in Maharashtra (India) and Panigrahi and Panigrahi (2015) for state of Odisha were also of the same opinion. Women walk for an average of 6 km per day in Asia and Africa to collect water for drinking and domestic use by carrying loads over long distance as reported by Caruso

(2017). Adoption of roof water harvesting reduces drudgery faced by women in water collection especially in rural areas. Smet and Moriarty (2001) reported that time saved in collecting water for domestic use can be utilised for other productive purposes. UNICEF (2013) showed the concern that health risks may arise by consuming contaminated water with infectious agents, toxic chemicals and radiological hazards. Improving access to safe drinking-water can result in tangible improvements to health.

Rooftop area available to collect the runoff from rainfall was calculated using GIS for community in villages or city, industry as well as institutions throughout the world (Giridhar *et al,.* 2013; Pawar and Mali, 2013; Al-Houri *et al.,* 2014; Nthuni *et al.,* 2014; Taher, 2014 and Shinde and Gaikwad, 2016). Water available from rooftop highly depends on rainfall of a particular area. As rainfall is highly uncertain, "design rainfall" *i.e.* rainfall amount which will serve as an input for calculating the water harvesting potential can be estimated based on past rainfall records of the area. To plan about the roof water harvesting project with its probable benefit and economics, the potential of roof water harvesting is to be estimated. Abdul *et al.* (2009) and Ndiritu *et al. (*2014) used Weibull plotting position formula while Pande and Telang (2014) used mean annual rainfall to estimate amount of water available for roof water harvesting. Madara *et al*. (2016) analysed the rooftop rainwater harvesting using average monthly rainfall as well as rainfall at probability of 66% as design rainfall. The demand of water for various kinds of uses which can be fulfilled from harvested roof water as well as design aspects can be evaluated by demand supply analysis. Study of roof water harvesting potential of a particular area based on GIS and demand supply analysis helps in decision making regarding its feasibility of implementation for individual or community. Such study demonstrates the planners as well as users its benefits and economics.

The present study has been planned with an objective to demonstrate GIS based planning for rooftop rainwater harvesting which will be useful not only for the study area but can also be replicated by researchers and planners for other areas.

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MATERIALS AND METHODS

The present study was undertaken for Junagadh Agricultural University campus, Junagadh (Coordinates 21° 30' 55.69'' N latitude, 70° 27' 23.19'' E longitude) which falls in semiarid region at southwest of Gujarat with 900 mm average annual rainfall. The availability of runoff from rooftops of campus buildings was estimated using two design rainfalls (i) rainfall at 66% probability of exceedance using Weibull's formula and (ii) average monthly rainfall. As more than 90% of rainfall of the study area occurs in June to September, the analysis was carried out by considering the monthly rainfall of these four months for the year 1961 to 2015.

The Weibull's equation was used to calculate the probability of exceedance. The rainfall depth at 66% probability of exceedance was obtained from graph of rainfall (mm) vs probability of exceedance (%). The Quantum GIS was used to calculate rooftop area of buildings. It is an open-source cross-platform desktop geographic information system (GIS) application that supports viewing, editing, and analysis of geospatial data. Dadhich and Pratishtha (2016) also used Google Earth and QGIS tool to convert raster file into vector file for estimating the area. Following were the steps to calculate the rooftop area using Google Earth and QGIS:

i) Digitize the polygons of rooftops of all buildings under the study area in Google Earth and export as .kmz file.

ii) Import file to QGIS by adding vector layer and convert .kmz raster file of google earth to vector file which results in vector map of the polygons of rooftop of all buildings.

iii) Area calculation is done by field calculator function of QGIS

iv) The Gould and Nissen (1999) formula was used to calculate harvesting potential for given design rainfall as below

$$Q = C \times A \times I \quad ...(1)$$

Where,
Q = Harvesting potential, litre
A = Area of rooftop catchment, (m^2)
I = Amount of rainfall (mm)
C = Runoff coefficient

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As most of the JAU campus buildings has cement concrete roofs, runoff coefficient was taken as 0.85 for this study as suggested by Rande (2000). Sample calculation of demand and supply analysis was done for four different categories of buildings existing in the campus i.e. College, Hostel, Multi storey office and residential quarters. The College of Agricultural Engineering and Technology (CAET) as College, Sardar Patel Hostel as Hostel building, University Bhavan (3 Storey office building) as Office and Panchvati Quarter-1 as residential building was selected for analysis. Per capita water requirement for college was taken as 45 lit/capita/day as per Anonymous (2010), Bureau of Indian Standards. For other three categories viz. hostel, office and residential building, demand was calculated based on various types of use of harvested water, as per Anonymous (2013), Ministry of Drinking Water & Sanitation, Government of India Guidelines. For hostel 18lit/capita/day, for multi storey office 13lit/capita/day and for residence 55 lit/capita/day was taken. In hydrological study "Water Year" is used to describe a time period of 12 months for which precipitation totals are measured. For the study area "Water Year" is period between June 1st of a calendar year and May 31st of next calendar year. The monthly demand was deducted from the monthly potential and balance brought forward from the previous month. From the analysis, percentage of water demand and day of water year upto which demand can be fulfilled from estimated roof water harvesting potential was found out. The size of storage tank was decided based on maximum amount of water available for storage. Water available for recharge per year was also calculated by deducting the total yearly demand from total potential (supply) from rooftop.

RESULTS AND DISCUSSION

Design Rainfall

The results of probability analysis of rainfall using Weibull's plotting position for month of June, July, August and September is given in Fig.1. Based on the probability analysis as well as average monthly rainfall, two design rainfall for each of the four month was obtained as given Fig. 1.

Total rainfall available for design rainfall (i) and design rainfall (ii) were 502.5 mm and 881 mm

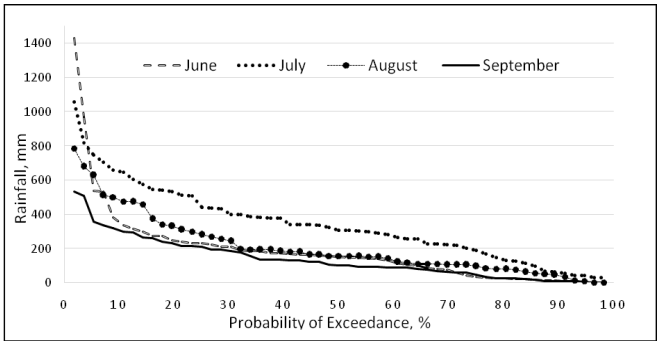


Fig. 1: Probability analysis of rainfall

respectively for harvesting the roof water. The roof water harvesting potential for July was highest for both the design rainfall. For design rainfall (i), share of July and August is 67% in the total available rainfall for harvesting. In case of design rainfall (ii), these two months contributed to 62% in the total available rainfall. This result suggested that July and August are very crucial for tapping the maximum rainwater for harvesting.

Roof Water Harvesting Potential Using GIS

Shinde and Gaikwad (2016) stated that it is very tedious work to assess the catchments available for rooftop rainwater harvesting. They employed GIS technique to calculate the area of various types of roofs for the measurement of its potential and planning for the area under study. The rooftop areas of all 81 buildings of JAU campus which included buildings such as colleges, hostels, offices and residences were digitized in Google Earth (image date 20 September 2016) which is shown in Fig. 2. The total rooftop area of campus buildings was

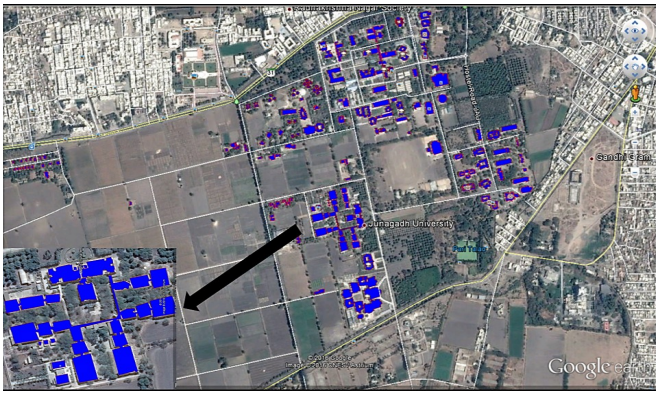


Fig. 2: Digitized rooftop areas of JAU campus buildings

found as 1,04,012 m^2. Out of total 81 campus buildings, rooftop areas of some buildings in various categories were measured manually to check the accuracy of procedure adopted in the study. The accuracy in measuring the rooftop area was found about 95% which shows that the procedure followed in the study to calculate the rooftop using Google Earth and QGIS is quite reliable.

The roof water harvesting potential for each building was calculated using (eq. 1). Out of total 81 buildings of the campus, smallest building has rooftop area of 88 m^2 while largest building has rooftop area of 14844 m^2. Considering these two buildings, roof water harvesting potential of campus buildings ranged from minimum 37.59 kiloliter to maximum 6340.24 kiloliter per year for design rainfall (i). For design rainfall (ii), it ranged from minimum 65.90 kiloliter to maximum 11115.93 kilo liter per year. The roof water harvesting potential of the campus is given in Table 1.

Demand Supply Analysis

The demand according to various sources of use was calculated as per the norms mentioned earlier. It was assumed that the harvested rainwater will be used to fulfill the demand of source as mentioned in Table 2. The demand – supply Analysis of college based on design rainfall as 66% probability of exceedance and monthly average rainfall is shown in Table 3 and Table 4, respectively. The cumulative demand and cumulative supply at various days of a water year is shown in Fig 3. Roof water harvesting potential of 6340.24 kiloliter for college can fulfill demand till 281st day of water year i.e. upto 11th March for 66% probability of exceedance design rainfall with maximum storage required is 3640.24 kiloliter. For monthly average rainfall, harvesting potential of 11115.93 kiloliter can meet the demand throughout the year and maximum storage required is 8415.93 kiloliter.

Similarly, the analysis of demand supply was done for hostel, office and residence, the results of analysis are shown in Figs. 4, 5 and 6 respectively. The analysis shows that for available rooftop catchments and design rainfall as 66% probably of exceedance, 78%, 88% and 59% of respective yearly water demand can be fulfilled for college, hostel and multi storey office building. It suggests that 22%, 12% and 41% demand is required to be fulfilled by other source. While for residence, demand of entire year can be fulfilled and additionally, 26.58 kiloliter of water will be available for recharge or use for other purposes. Considering design rainfall as average monthly rainfall, the demand of all four categories of buildings can be fulfilled throughout the year and additionally 3015.93, 293.04, 27.01 and 121.171 kiloliter water will be available for recharge.

Table 1. Roof water harvesting potential of Junagadh Agricultural University campus

Month	June	July	August	September	Total
For Design rainfall (i), kl	7426.46	20245.94	9813.53	6940.2	44426.13
For Design rainfall (ii) kl	16797.94	30236.29	18566.14	12289.02	77889.39

Table 2. Monthly water demand for various buildings of Junagadh Agricultural University campus

S. No.	Name of building	Demand (l/capita/day)	Source of demand	No. of capita	Monthly demand (kl)
1	College	45	Overall	500	675.0
2	Hostel	18	Drinking, cooking, washing utensils and premises	90	48.6
3	Multi storey office	13	Drinking, sanitation and miscellanies use	160	62.4
4	Residence	55	Drinking, cooking, bathing, washing utensils and house, toilets, washing of clothes and other uses	05	08.25

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Table 3. Demand – supply analysis for college at design rainfall (i)

Month	Monthly supply from rooftop (kl)	Day of water year	Cumulative demand, (kl)	Cumulative supply (kl)	Water available at the end of month (kl)
June	1059.86	30	675	1059.86	384.86
July	2889.38	61	1350	3949.24	2599.24
August	1400.53	91	2025	5349.77	3324.77
September	990.47	122	2700	6340.24	3640.24
October	0	153	3375	6340.24	2965.24
November	0	183	4050	6340.24	2290.24
December	0	214	4725	6340.24	1615.24
January	0	245	5400	6340.24	940.24
February	0	273	6075	6340.24	265.24
March	0	304	6750	6340.24	0
April	0	334	7425	6340.24	0
May	0	365	8100	6340.24	0

Table 4. Demand – supply analysis for college at design rainfall (ii)

Month	Monthly supply from rooftop (kl)	Day of water year	Cumulative demand, (kl)	Cumulative supply (kl)	Water available at the end of month (kl)
June	2397.31	30	675	2397.31	1722.31
July	4315.15	61	1350	6712.46	5362.46
August	2649.65	91	2025	9362.11	7337.11
September	1753.82	122	2700	11115.93	8415.93
October	0	153	3375	11115.93	7740.93
November	0	183	4050	11115.93	7065.93
December	0	214	4725	11115.93	6390.93
January	0	245	5400	11115.93	5715.93
February	0	273	6075	11115.93	5040.93
March	0	304	6750	11115.93	4365.93
April	0	334	7425	11115.93	3690.93
May	0	365	8100	11115.93	3015.93

Table 5. Abstract of demand supply analysis for design rainfall (i)

S. No.	Name of building	Roof- top area (m^2)	Total yearly demand (kl)	Total supply (kl)	Day of water year upto which demand can be fulfilled	Maximum storage required (kl)	% of yearly demand that can be fulfilled	Water available for recharge (kl)
1	College	14844	8100.00	6340.2	281	3640.24	78	-
2	Hostel	1110	538.20	474.10	292	297.70	88	-
3	Multi storey office	1036	748.80	442.51	212	192.91	59	-
4	Residence	294	99.00	125.58	Throughout year	92.58	100	26.58

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Table 6. Abstract of demand supply analysis design rainfall (ii)

S. No.	Name of building	Roof-top area (m²)	Total yearly demand (kl)	Total supply (kl)	Day of water year upto which demand can be fulfilled	Maximum storage required, (kl)	Water available for recharge (kl)
1	College	14844	8100.00	11115.93	Throughout year	8415.93	3015.93
2	Hostel	1110	538.20	831.24		636.84	293.04
3	Multi storey office	1036	748.80	775.81		526.21	27.01
4	Residence	294	99.00	220.17		187.17	121.17

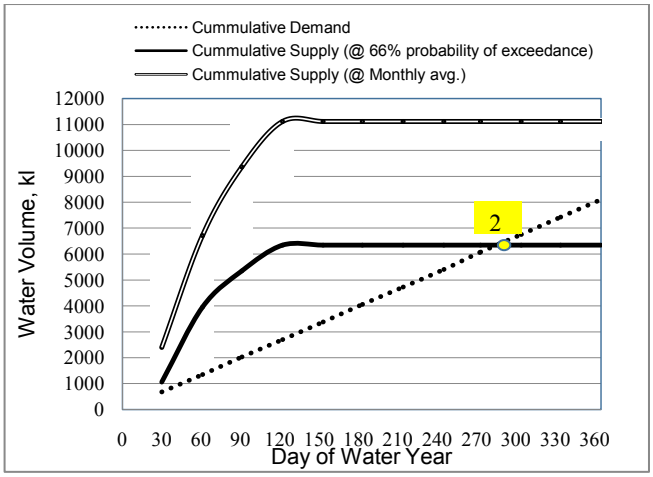


Fig. 3: Demand – supply analysis of college

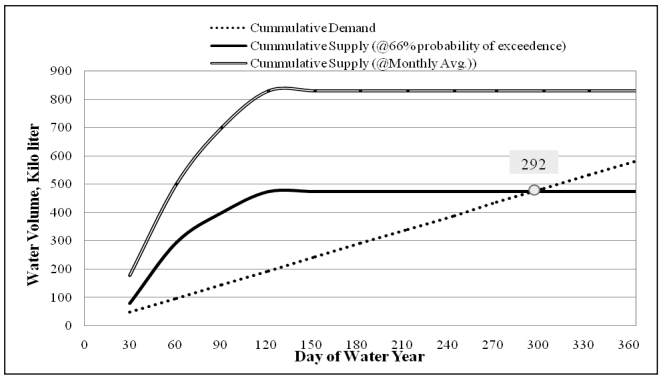


Fig. 4: Demand – supply analysis of hostel

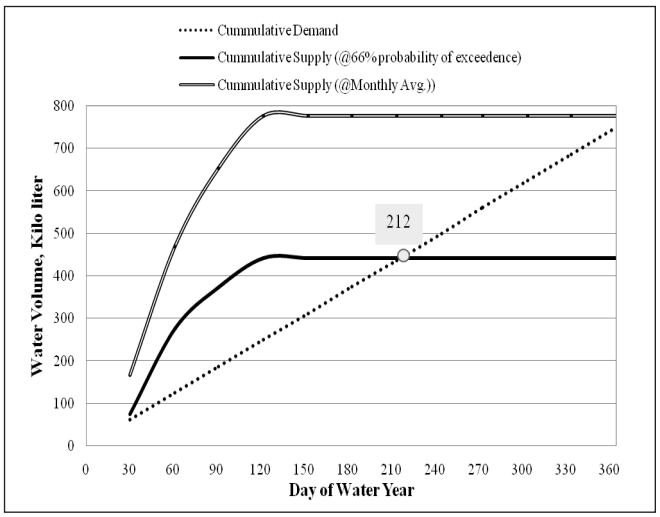


Fig. 5: Demand – supply analysis of multi storey office

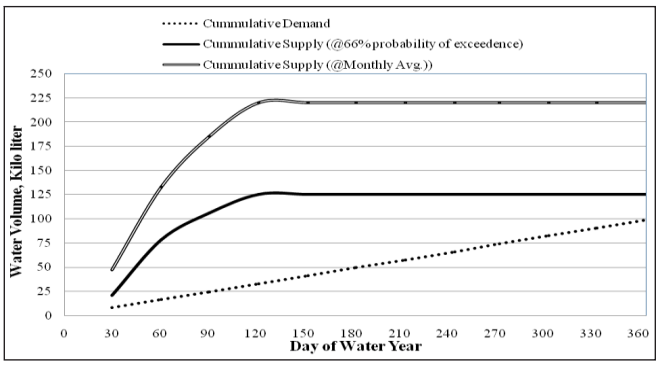


Fig. 6: Demand – supply analysis of residence

If water is to be utilized by storing in storage tank, the size of storage tank for each building will be equal to maximum storage required. The present study reveals that size of storage tank required for design rainfall (i) is 57%, 53%, 63% and 51% smaller for college, hostel, office and residence respectively as compared to that of design rainfall (ii). Kulkarni (2016) showed the concern that initial investment becomes limiting factor in adopting this technology. Determining the size of storage tank required is most critical as storage tank contributes to about 80% of the total cost of the system. Smet and Moriarty (2001), Liaw and Tsai (2004) and Kahinda *et al.* (2007) were also of the opinion that the storage tank accounts for the highest cost for a rooftop rainwater harvesting system and installation of correct size of tank is very crucial for design. Ward *et al.* (2010) observed that over

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sizing tanks can lead to excessive system capital costs, which currently hinder the uptake of systems. Considering the various studies mentioned above, design rainfall for the roof water harvesting system can be decided based on purpose of use. If cost is a limiting factor for the adoption, the design of rooftop rainwater harvesting should be done based on at 66% probability of exceedance rainfall and harvested water should be restricted for important purposes so that entire demand can be fulfilled. If cost is not limiting factor and ground water recharge is also a priority, the structure should be designed at average rainfall.

Evaluating the quantity of water which can be harvested from available roof area based on rainfall analysis for a given locality using GIS tools along with study of demand supply analysis will help community as well as policy makers decision to promote the roof water harvesting technique. The idea of amount of water that could be harvested and demand supply analysis can serve as an input for estimation of cost of the project and benefits.

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CONCLUSIONS

The rooftop rainwater harvesting potential at design rainfall as 66% probability of rainfall can fulfill the demand of college, hostel and multi storey office up to 281st day, 292nd day and 212th day of water year. For design rainfall at monthly average, the demand of entire year can be fulfilled from rooftop rainwater harvesting with 3015.93, 293.04, 27.01 and 121.171 kiloliter excess water are available for recharge. If initial investment is a limiting factor for adoption of rooftop rainwater harvesting system, the storage tank should be designed at 66% probability of design rainfall for saving cost on storage structure.

REFERENCES

Abdul H A; Ahmed A A; Shaqlof H. 2009. Assessment the quality and quantity of harvested rainfall from different catchments systems study case: North West of Libya. Thirteenth International Water Technology Conference, IWTC 13 , Hurghada, Egypt.

Adriano T E; Antonio E M; Rodolfo G N. 2011. Rainwater harvesting, quality assessment and utilization in region I. International Scientific Research Journal, 3(2):145-155.

Agarwal R; Garg S; Kaur S P; Kaushal M P; Biwalkar N. 2010. Impact of rooftop rainwater harvesting for groundwater recharge at Ludhiana. Journal of Agricultural Engineering, 47(1):46-48.

Alam R G; Munna M A; Chowdhury M S; Sarkar K A; Ahmed M P; Rahman M T; Jesmin F A; Toimoor M A. 2012. Feasibility study of rainwater harvesting system in Sylhet city - Environmental Monitoring and Assessment, 184(1):573-580.

Al-Houri Z M; Abu-Hadba O K; Hamdan K A. 2014. The potential of rooftop rainwater harvesting as a water resource in Jordan: featuring two application case studies. International Journal of Environmental, Chemical, Ecological, Geological and Geophysical Engineering, 8(2):147-153.

Anonymous. 2010. Indian standard code of basic requirements for water supply, drainage and sanitation (fourth revision). Bureau of Indian Standards. pp 5.

Anonymous. 2013. National rural drinking water programme movement towards ensuring people's drinking, water security in rural India, Guidelines – 2013. Ministry of Drinking Water & Sanitation, Government of India.

Caruso B. 2017. Women still carry most of the world's water. https://theconversation.com/ women-still-carry-most-of-the-worlds-water-81054.

Dadhich G; Pratishtha M. 2016. GIS based analysis for rooftop rainwater harvesting. International Journal of Computer Science & Engineering Technology, 7(4):129-143.

Giridhar M V; Chandra A S; Viswanath G K. 2013. Identification of suitable locations for rooftop rainwater harvesting structures. International Journal of Applied Sciences and Engineering Research, 2(2):101-106.

Gould J; Nissen P E. 1999. Rainwater catchment systems for domestic supply: design, construction and implementation, London. I.T. Publications.

Jankar P D; Bhanuse M M. 2013. Domestic rooftop rainwater harvesting - a case study of village. International Journal of Research in Engineering and Technology, 2(12):50-52.

Kulkarni S J. 2016. Review on studies, research and surveys on rainwater harvesting. International Journal of Research and Review, 3 (9): 1-11.

Liaw C H; Tsai Y L. 2004. Optimum storage volume of rooftop rainwater harvesting systems for domestic

use. American Water Resources Association, 40 (4): 901-912.

Kahinda J M; Akpofure E T; Boroto, J R. 2007. Domestic rainwater harvesting to improve water supply in rural South Africa. Physics and Chemistry of the Earth, 32:1050-1057.

Madara D S; Namango S S; Katana H. 2016. Potential of roof rain water harvesting at an industrial setup. Journal of Environment and Earth Science, 6(7):110-117.

Ndiritu J G; Mccarthy S; Tshirangwana N. 2014. Probabilistic assessment of the rainwater harvesting potential of schools in South Africa. *In*. Proceedings of 6th International Symposium on Integrated Water Resources Management held in Bologna, Italy, 4–6 June, pp 435-440.

Nthuni S M; Lubker T; Schaab G. 2014.Modelling the potential of rainwater harvesting in western Kenya using remote sensing and GIS techniques. South African Journal of Geomatics, 3(3):285-301.

Paul, J C; Panigrahi, B. 2016. Artificial conservation measures on groundwater recharge, irrigation potential and productivity of crops of Bharkatia Watershed, Odisha, Journal of Soil and Water Conservation, Vol. 15(2): 1-7.

Pande, P; Telang, S. 2014.Calculation of rainwater harvesting potential by using mean annual rainfall, surface runoff and catchment area. Global Advanced Research Journal of Agricultural Science, 3(7): 200-204.

Panigrahi, B; Panigrahi, K. 2015. Design of rooftop rainwater harvesting structure for domestic water supply in Bhubaneswar. Journal of Geosciences Research, Vol. 1(2): 137-142.

Pawar V S; Mali S P. 2013. Potential roof rainwater harvesting in Pirwadi village of Kolhapur district, Maharashtra (India) - A Geospatial Approach. Journal of Research in Humanities and Social Science, 1(4):19-24.

Rande R. 2000. A water harvesting manual for urban areas: Case studies from Delhi -Publication Center for Science and Environment, New Delhi.

Shinde S D; Gaikwad V P. 2016. Application of GIS for mapping rainwater harvesting potential: a case study of Nidhal village in Satara district, Maharashtra, India. International Journal of Research in Applied, Natural and Social Sciences, 4(5):141-148.

Smet J; Moriarty P. 2001. DGIS Policy Supporting Paper: Rooftop Rainwater Harvesting. IRC Delft, the Netherlands. pp 19.

Taher T M. 2014. Quantity and quality considerations of rooftop rainwater harvesting as a substantial resource to face water supply shortages. International Journal of Water Resources and Arid Environments, 3(1):1-10.

UNICEF. 2013. Progress on drinking water and sanitation 2012, World Health Organization update. New York.

Ward S; Memon F A; Butler D. 2010. Rainwater harvesting: model-based design evaluation. Centre for water systems, School of Engineering, Computing and Mathematics, University of Exeter. Water Science and Technology, 61(1):85-96.

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